

MACRONUTRIENTS-PROTEIN

Proteins are high-molecular-weight nitrogenous biomolecules composed of one or more chains of amino acids joined by peptide bonds.

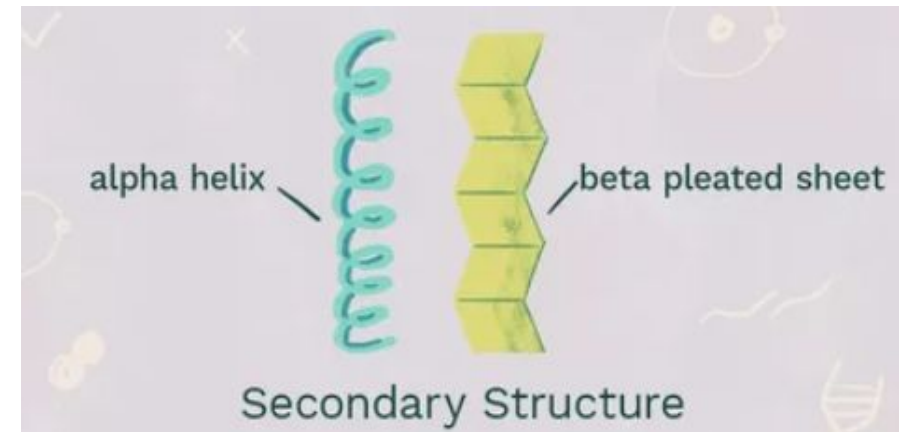
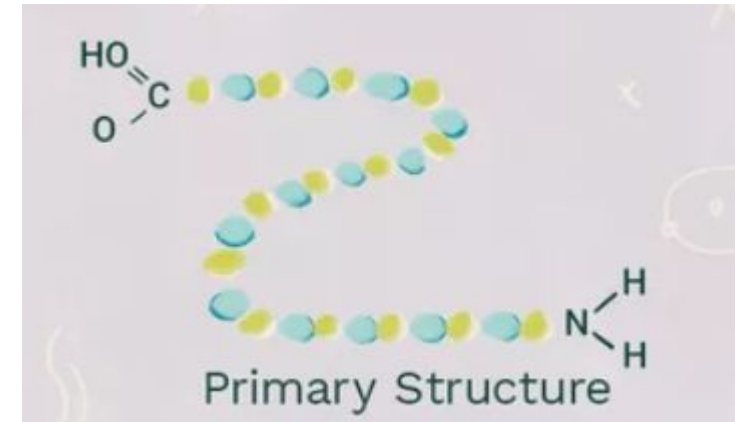
CLASSIFICATION OF PROTEINS

There are many proteins in the human body, and each one has a very specialized function.

1. Classification of Proteins Based on Structure

- **Primary Structure** – This is the basic sequence of amino acids in a chain. The behavior of the protein can be changed by even minor modifications to this region. For example, a change in hemoglobin's primary structure leads to sickle cell anemia.

- **Secondary Structure** – At this level, the protein chain forms simple patterns such as alpha-helices or beta-sheets, held together by hydrogen bonds. These patterns add stability.

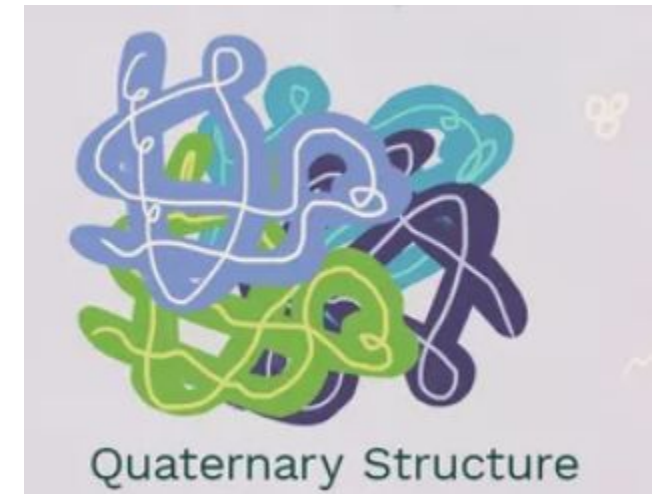


CLASSIFICATION OF PROTEINS

- **Tertiary Structure** – The entire chain folds into a unique three-dimensional shape. This folding is extremely important because the protein's function depends on it. Enzymes, for instance, rely on their tertiary structure to form active sites where reactions take place.



- **Quaternary Structure** – Some proteins consist of multiple chains or subunits that work together. Hemoglobin is a classic example because it has four subunits that together bind oxygen efficiently.



CLASSIFICATION OF PROTEINS

2. Classification of Proteins Based on Chemical Composition

Proteins can also be grouped depending on what they are made of:

- **Simple Proteins** – These consist only of amino acids. When broken down, they give only amino acids. Examples include albumin (found in egg white and blood plasma) and globulins (important in immunity).
- **Conjugated Proteins** – These are prosthetic groups—proteins joined with non-protein components. Examples are:
 - **Hemoproteins** such as hemoglobin (with a heme group).
 - **Lipoproteins** (proteins with lipids, important for cholesterol transport).
 - **Glycoproteins** (proteins linked with carbohydrates, important in cell recognition).
- **Derived Proteins** – These are formed when simple or conjugated proteins undergo breakdown or chemical modification.

Examples are peptones and proteoses, formed during digestion.

CLASSIFICATION OF PROTEINS

3. Classification of Proteins Based on Shape

- Proteins also differ in shape, and this influences their role in the body. There are mainly two categories:
- **Fibrous Proteins** – These are long, thread-like, and insoluble in water. They mostly provide mechanical support and strength. Examples include collagen (in skin and tendons), keratin (in hair and nails), and elastin (in ligaments).
- **Globular Proteins** – These are compact, rounded, and generally soluble in water. They are functional proteins involved in metabolism, transport, and defense. Examples include enzymes, hormones like insulin, and antibodies.
- Fibrous proteins mainly give structure, while globular proteins are more active in biological processes.

CLASSIFICATION OF PROTEINS

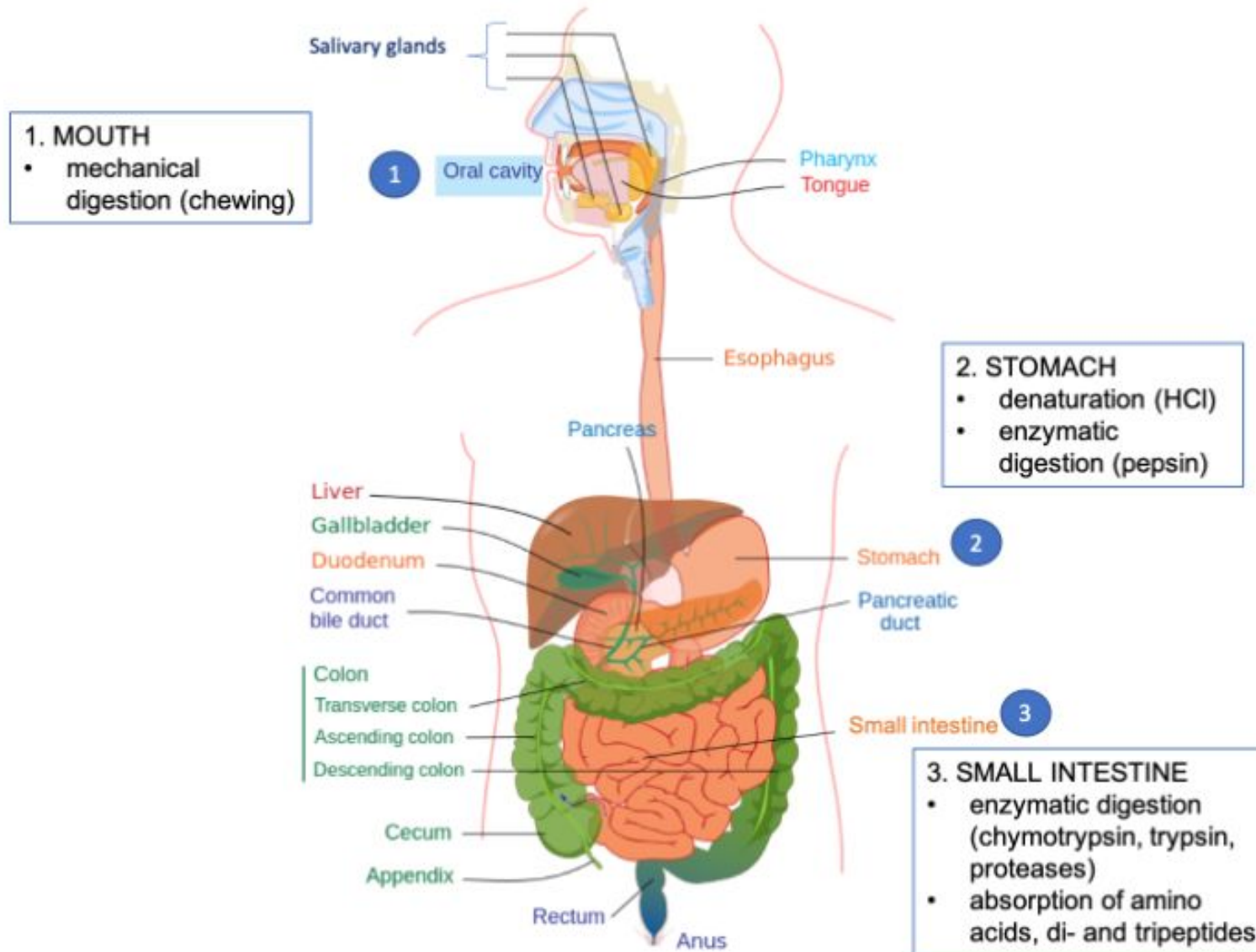
4. Classification of Proteins Based on Function

- The most practical way of classifying proteins is by their function. This system helps us understand what role each protein plays in life processes:
- **Enzymes** – Proteins that accelerate biochemical reactions. Example: lipase for fat breakdown.
- **Transport Proteins** – Carry molecules like oxygen and nutrients. Example: hemoglobin for oxygen transport.
- **Defensive Proteins** – Act as the body's soldiers. Example: antibodies against infections.
- **Structural Proteins** – Provide shape and support. Example: collagen in connective tissues.
- **Storage Proteins** – Store essential elements. Example: ferritin for iron storage.
- **Hormonal Proteins** – Act as messengers to regulate body processes. Example: insulin controlling blood sugar.
- **Contractile Proteins** – Help in movement. Example: actin and myosin in muscles.
- By dividing proteins in this way, biochemistry explains how proteins cover every aspect of life, from energy regulation to disease defense.

IMPORTANCE OF PROTEINS

- They are **essential for growth and repair** of body tissues.
- Proteins act as **catalysts** in almost all chemical reactions.
- They are crucial for **immunity**, since antibodies are proteins.
- Many proteins act as **carriers**, transporting gases, hormones, and nutrients.
- They also serve as **chemical messengers** to regulate various functions.
- In times of starvation, proteins may also act as an **energy source**.

PROTEIN DIGESTION AND ABSORPTION

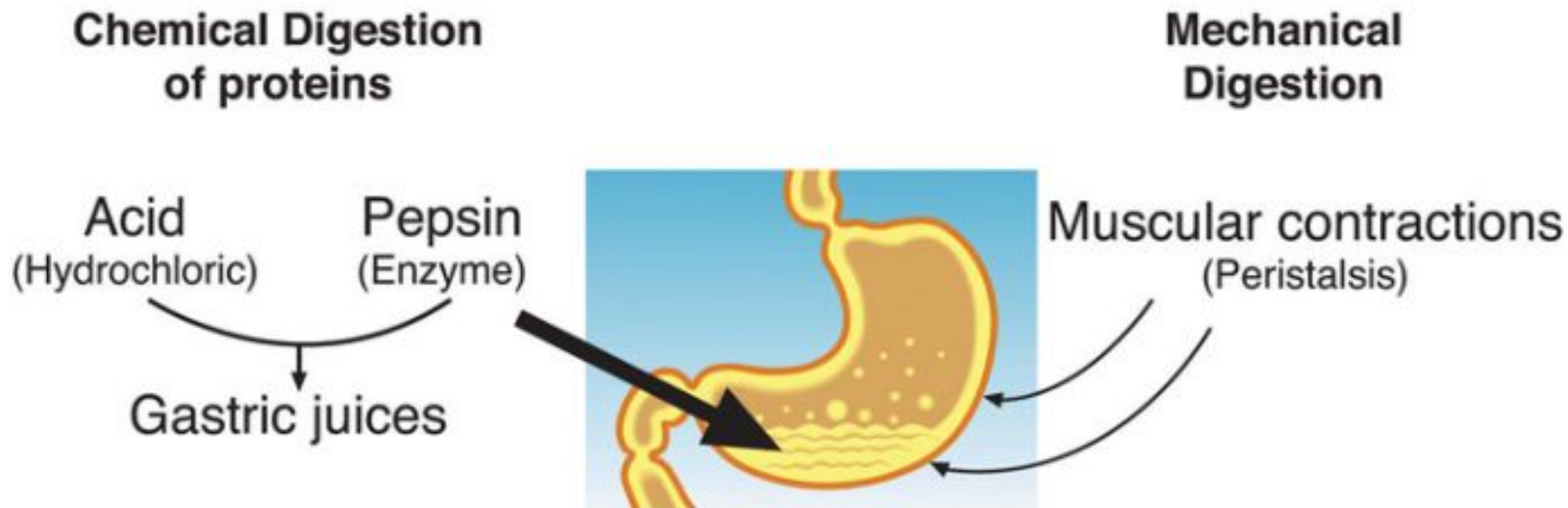


PROTEIN DIGESTION AND ABSORPTION

1. Protein digestion in the mouth

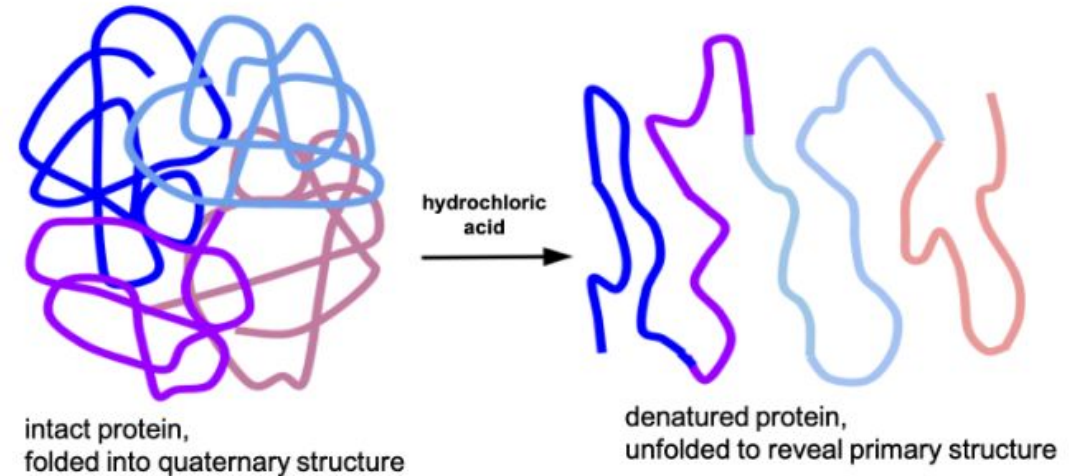
The teeth begin the **mechanical breakdown** of large pieces into smaller pieces that can be swallowed. The **salivary glands secrete saliva** to aid swallowing and the passage of the partially mashed food through the esophagus.

2. Protein digestion in the stomach

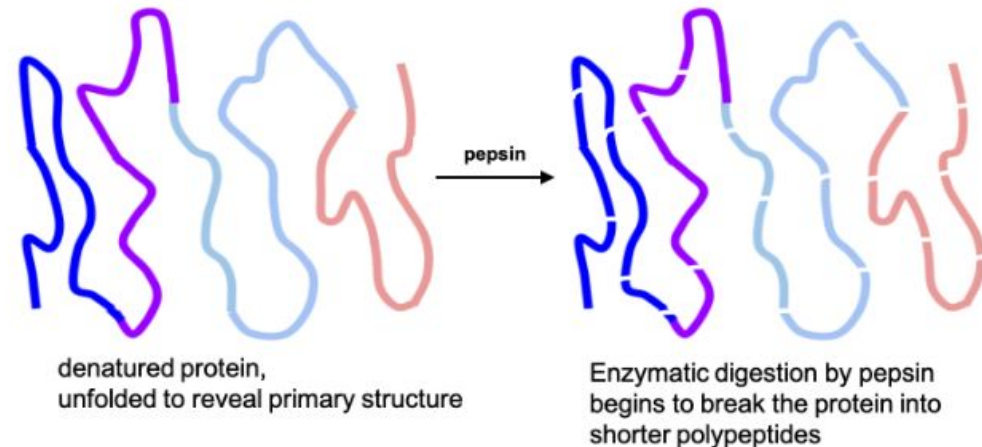


PROTEIN DIGESTION AND ABSORPTION

- proteins are denatured because of the acidity of hydrochloric acid.

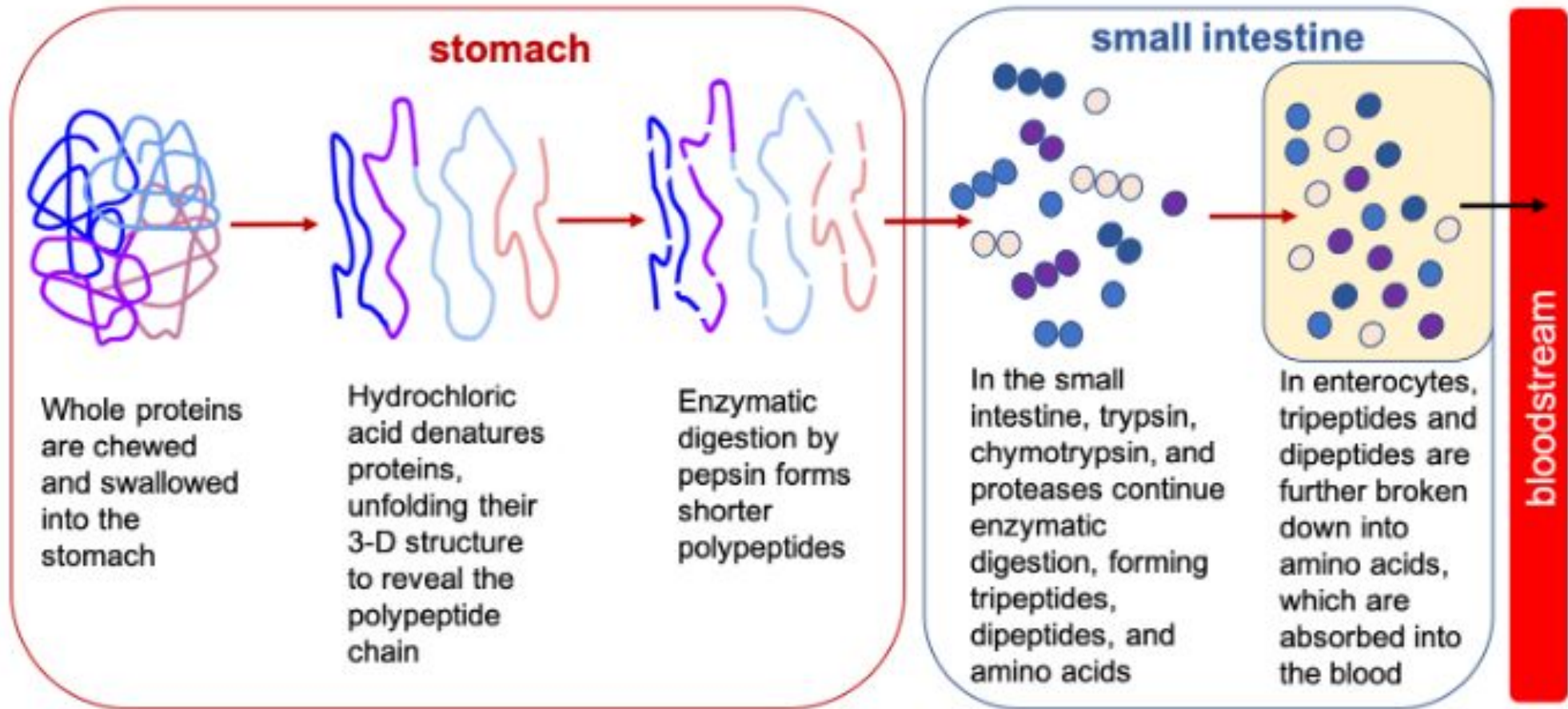


- Enzymatic digestion of proteins begins in the stomach with the action of the enzyme pepsin.



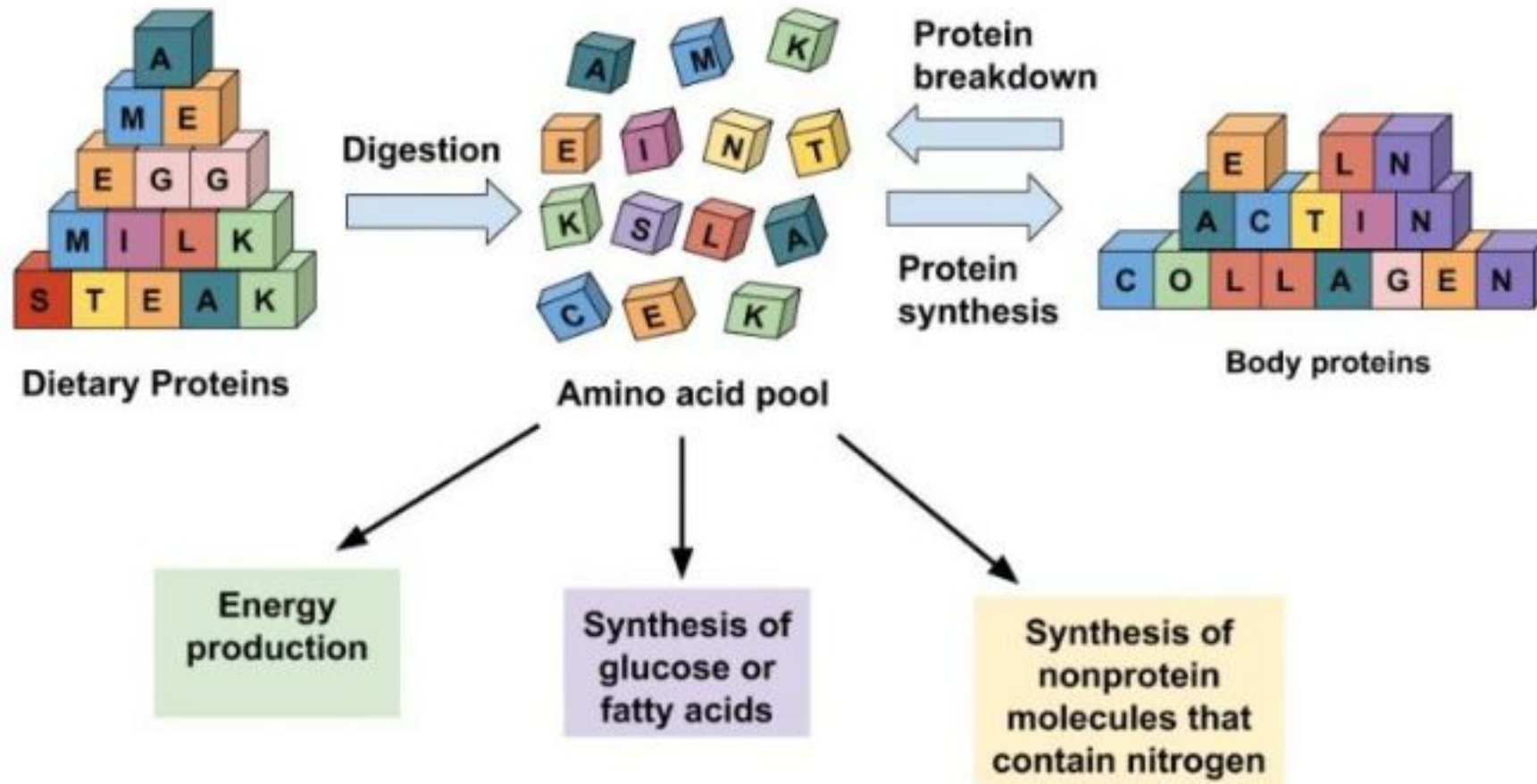
PROTEIN DIGESTION AND ABSORPTION

- Protein digestion and absorption in the small intestine

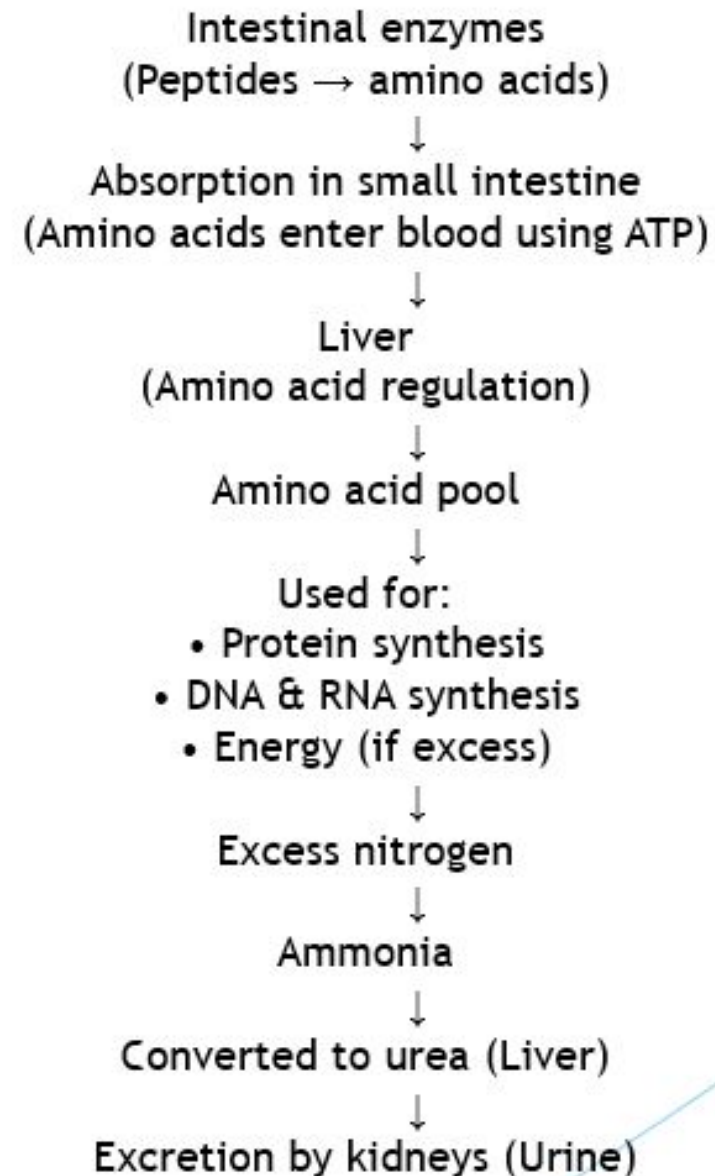
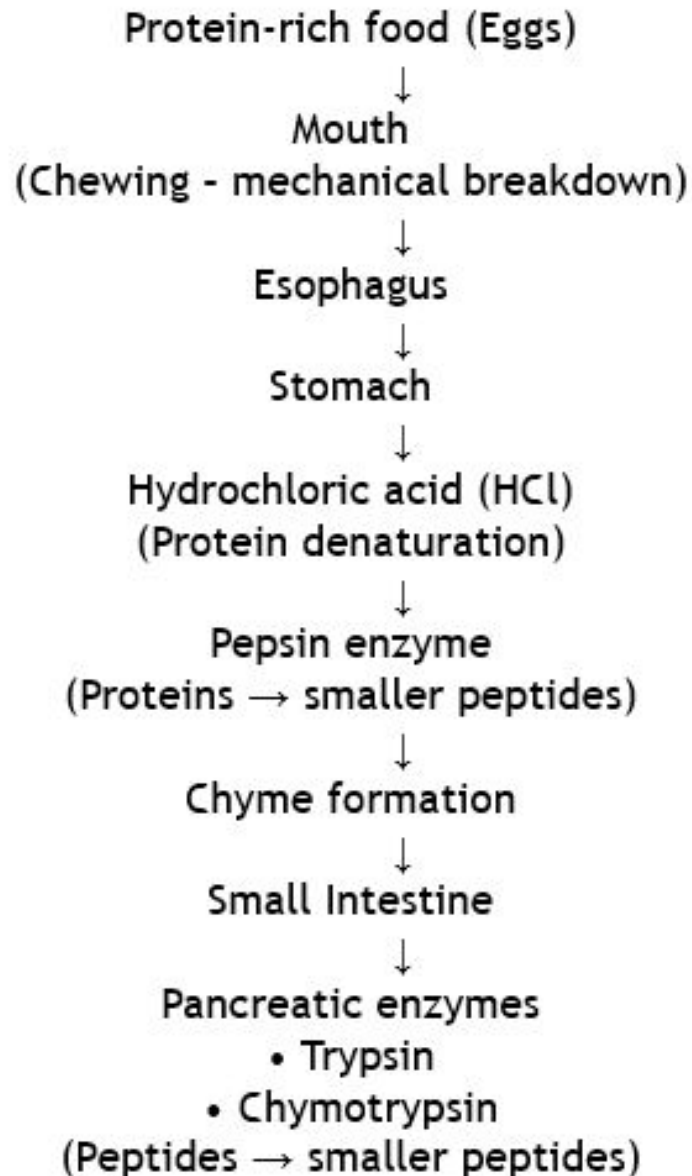


FATE OF ABSORBED AMINO ACIDS

- Absorbed amino acids



PROTEIN DIGESTION AND ABSORPTION



RDA OF PROTEINS

Recommended Dietary Allowances (RDA) for protein in India, based on the Indian Council of Medical Research – National Institute of Nutrition (ICMR-NIN) guidelines.

Adults (Healthy Individuals)

- RDA for adults (average healthy man/woman):
0.83 grams of protein per kilogram of body weight per day.

This means roughly:

- ~54 g/day for an average adult man
- ~46 g/day for an average adult woman

Children & Adolescents

- Protein needs vary with age (values approximate):
- **1–3 years:** ~13 g/day
- **4–8 years:** ~19 g/day
- **9–13 years:** ~34 g/day
- **14–18 years:** ~46–52 g/day (girls/boys)

FUNCTIONS OF PROTEINS

1. **Body building and growth**

- Proteins form the **structure of cells and tissues**.
- Essential for **growth** in children and **tissue repair** in adults.

2. **Repair and maintenance of tissues**

- Repairs **worn-out tissues**, wounds, and damaged cells.
- Maintains **skin, muscles, organs, hair, and nails**.

3. **Formation of enzymes**

- All **enzymes are proteins**.
- Enzymes speed up **digestion and metabolic reactions**.

4. **Formation of hormones**

- Some hormones are protein in nature (e.g., **insulin, growth hormone**).
- Regulate **body functions and metabolism**.

FUNCTIONS OF PROTEINS

5. Immunity (antibodies)

- Proteins form **antibodies** that protect against infections and diseases.

6. Transport of substances

- Transport proteins carry materials in blood:
 - **Hemoglobin** carries oxygen
 - **Albumin** carries nutrients, drugs, hormones

7. Maintenance of fluid balance

- Maintain **osmotic pressure** and prevent edema.
- Help keep proper **water balance** in blood and tissues.

8. Maintenance of acid–base balance

- Proteins act as **buffers** to maintain normal pH of blood.

9. Energy source (when needed)

- Provide **4 kcal/g**.
- Used for energy when carbohydrates and fats are insufficient.

SOURCES OF PROTEINS

- Animal Sources- Milk and Milk products, Egg, Fish, Chicken, Mutton, Beef, Pork.
- Plant Sources- Cereals, Pulses and Legumes, Nuts and Oilseeds, Soybean.

DEFICIENCY OF PROTEINS

When the diet lacks enough protein for a long time, the body cannot grow, repair, or protect itself properly.

In children

1) Kwashiorkor (protein deficiency with adequate calories)

- Edema (swelling of legs, face)
- Moon face appearance
- Hair discoloration (reddish/brown), sparse hair
- Pot belly
- Skin lesions (“flaky paint” dermatosis)
- Poor growth and irritability

DEFICIENCY OF PROTEINS

2) Marasmus (deficiency of protein and calories)

- Severe wasting of muscles and fat
- Very thin body (“skin and bones”)
- Sunken eyes, old man look
- No edema
- Stunted growth

In Adults

- Muscle wasting and weakness
- Weight loss
- Fatigue
- Frequent infections (low immunity)
- Slow wound healing
- Hair fall, brittle nails
- Edema in severe cases